Practical 09 :

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code :

```
  GNU nano 8.7.1
#include <stdio.h>

int main() {
    int pages[12] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frame[3];
    int faults = 0, hits = 0;
    int i,j,k,pos;

    for(i=0;i<3;i++)
        frame[i] = -1;

    for(i=0;i<12;i++){
        int found = 0;

        for(j=0;j<3;j++){
            if(frame[j] == pages[i]){
                hits++;
                found = 1;
                break;
            }
        }

        if(!found){
            int farthest = -1;
            pos = -1;

            for(j=0;j<3;j++){
                int next = -1;
                for(k=i+1;k<12;k++){
                    if(frame[j] == pages[k]){
                        next = k;
                        break;
                    }
                }

                if(next == -1){
                    pos = j;
                    break;
                }

                if(next > farthest){
                    farthest = next;
                    pos = j;
                }
            }

            frame[pos] = pages[i];
```

```
                pos = j;
            }
        }

        frame[pos] = pages[i];
        faults++;
    }
}

printf("Page Hits = %d\n", hits);
printf("Page Faults = %d\n", faults);
printf("Hit Percentage = %.2f%%\n", (hits/12.0)*100);
printf("Fault Percentage = %.2f%%\n", (faults/12.0)*100);

return 0;
```

Output :

```
piyus@piyush MINGW64 ~
$ nano practical09.c

piyus@piyush MINGW64 ~
$ gcc practical09.c -o practical09

piyus@piyush MINGW64 ~
$ ./practical09
Optimal Page Replacement Algorithm
Page References: 1 2 3 4 1 2 5 1 2 3 4 5
No of Frames: 3

Incoming       Frame1  Frame2  Frame3  Page Fault
1              1       -       -       Fault
2              1       2       -       Fault
3              1       2       3       Fault
4              1       2       4       Fault
1              1       2       4       Hit
2              1       2       4       Hit
5              1       2       5       Fault
1              1       2       5       Hit
2              1       2       5       Hit
3              3       2       5       Fault
4              4       2       5       Fault
5              4       2       5       Hit

Total Page References: 12
Page Hits: 5
Page Faults: 7
Hit Percentage: 41.67
Fault Percentage: 58.33

piyus@piyush MINGW64 ~
$
```